Bayesian ARX Identification for Robust Control, Uncertainty Quantification, and Attack-Resilient Analysis

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***Abstract*— Bayesian system identification is attractive for control not only because it regularizes estimation, but because it preserves uncertainty in the objects that actually matter for design: poles, frequency response, closed-loop transients, and robustness margins. This paper presents a compact Bayesian Autoregressive with Exogenous Inputs (ARX) workflow and argues that such methods should be used more broadly in robust control and security-sensitive system identification, especially in weak-data applications such as automated model learning. The current implementation supports closed-form Bayesian and least-squares ARX fitting, isotropic and diagonal Gaussian priors, unknown-noise variants, posterior predictive inference, posterior-sampled stability analysis, frequency-response envelopes, Nyquist uncertainty plots, and closed-loop Monte Carlo propagation. These outputs expose where nominal least-squares conclusions are brittle and motivate two immediate control problems: posterior-risk controller synthesis and posterior-wide stealth destabilization through a Doyle–Glover–Khargonekar–Francis style Δ-feedback interconnection. The goal is to provide a technically concrete bridge from Bayesian identification to robust control and, ultimately, to Bayesian attack mitigation.**[1]

I. Introduction

Controllers are designed from models, but those models are learned from finite data. In practice, the identification stage is often the weakest link in the control pipeline: a nominal fit can appear persuasive even when the data are too short, too noisy, too weakly exciting, or too coarsely sampled to support the dynamical claims later made from that fit. This matters most when the downstream question is not only one-step prediction, but stability, frequency-domain robustness, closed-loop performance, or vulnerability of a feedback loop to malicious perturbations [1], [2].

A practical bottleneck in critical-infrastructure security motivates this work: many smaller utilities need cyber-physical models for threat analysis, resilience assessment, and mitigation planning, but they often do not have the budget, personnel, or time to build validated models through a traditional, fully manual engineering process [3], [4]. Operational-technology systems are topology-dependent, highly interconnected, and frequently analyzed under limited or imperfect data, so a more realistic goal is an automated pipeline that starts from the network structure, telemetry, and operational records a utility already has. In that setting, Bayesian system identification is attractive because it avoids collapsing uncertainty into a single nominal model. Instead, it identifies which inferred dynamics are genuinely supported by the available data, highlights where the data are insufficient, and still enables downstream stability, robustness, and attack-impact analysis with explicit confidence statements rather than implicit overconfidence. This makes the Bayesian workflow a natural fit for low-resource critical-infrastructure settings, where analysis must proceed under incomplete information rather than ideal laboratory data.

Bayesian identification provides this analysis by replacing a single estimate with a posterior distribution over plausible plants. Recent work in Bayesian and regularized system identification has shown that probabilistic formulations can substantially improve calibration and robustness, particularly in limited-data regimes [5], [6]. For controls, the key advantage is that one can propagate posterior uncertainty directly into pole locations, Bode or Nyquist geometry, stability probabilities, and closed-loop responses, rather than collapsing all uncertainty into a single nominal model. Even in robust control theory, where system uncertainty is often modeled as a ball in Hilbert space centered around a nominal plant, using instead the Bayesian posterior as the uncertainty set is advantageous, as the posterior uncertainty set is directly connected to the actual system uncertainty inherited from the data, and is not merely an afterthought in which the engineer crosses their fingers hoping that the chosen uncertainty set is neither too conservative nor insufficient (or both, if the wrong distribution geometry is assumed).

We use a compact Bayesian ARX implementation [7] to advocate a broader methodological point: Bayesian system identification provides a natural, data-driven uncertainty representation for robust control and attack-resilient analysis. Concretely, the paper offers:

1) a Bayesian ARX workflow with isotropic and structured priors, known-noise and unknown-noise variants, and rolling order selection;
2) control-relevant uncertainty objects derived from posterior samples, including predictive bands, pole clouds, frequency envelopes, Nyquist uncertainty plots, and closed-loop response bands;
3) two regimes where the Bayesian viewpoint is especially important—underconstrained identification and sub-Nyquist sampling—because nominal fits can look crisp while the induced dynamics remain ambiguous;
4) a posterior-risk controller-design template and a posterior-wide Δ-feedback destabilization template that connect Bayesian identification to robust control and future attack mitigation.

[1]The text and codebase of this report were produced with the help of OpenAI's ChatGPT 5.3 Extended Reasoning model and Codex.

II. Bayesian ARX identification and prior structure

Consider the SISO discrete-time ARX model

$$y_t = \phi_t^\top \theta + e_t, \qquad e_t \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

with regressor

$$\phi_t = [y_{t-1}, \ldots, y_{t-n_a}, u_{t-1}, \ldots, u_{t-n_b}]^\top. \tag{2}$$

For stacked data $\mathcal{D} = (\Phi, y)$ and Gaussian prior

$$\theta \sim \mathcal{N}(\mu_0, \Sigma_0), \tag{3}$$

conjugacy yields

$$\Sigma_N^{-1} = \Sigma_0^{-1} + \frac{1}{\sigma^2}\Phi^\top\Phi, \qquad \mu_N = \Sigma_N\Big(\Sigma_0^{-1}\mu_0 + \frac{1}{\sigma^2}\Phi^\top y\Big). \tag{4}$$

The one-step predictive distribution at a new regressor $\phi_\star$ is

$$y_\star \mid \mathcal{D}, \phi_\star \sim \mathcal{N}\big(\phi_\star^\top \mu_N,\ \phi_\star^\top \Sigma_N \phi_\star + \sigma^2\big). \tag{5}$$

Equation (5) is the learned system model, equivalent to the system state equations $x_t = f(x_{t-1}, u_t, e_t)$, $y_t = g(x_t, u_t, e_t)$, and already exposes the central modeling benefit: observation noise and parameter uncertainty remain separate and can be propagated differently in downstream control tasks.

The present workflow emphasizes two prior families. The first is isotropic, $\Sigma_0 = \alpha^2 I$, which is useful when little directional prior information is available and one wants a simple regularizer that shrinks all coefficients symmetrically. The second is diagonal with distinct scales on autoregressive and input coefficients, which is often more appropriate when prior knowledge suggests that state-memory and actuation channels should be regularized differently. In short datasets, this choice is consequential: posterior covariance contracts only along directions excited by the data, so weak excitation leaves visibly different identifiable and nonidentifiable directions.

The implementation also includes an unknown-noise variant. Under a normal-inverse-gamma prior, the marginal predictive law becomes Student-t, which is helpful when small datasets or occasional outliers make Gaussian tails overconfident. Rolling-origin order selection for (n_a, n_b) is included so that structural uncertainty is not hidden inside a single in-sample order choice.

Before moving to the hard cases, it is useful to recall what the basic Bayesian predictive object means. Equation (5) produces a full one-step distribution rather than a single next-step estimate. This matters because a realized measurement can differ from the conditional mean without contradicting the model; what matters is whether it remains plausible under the predictive law. In short, our model only makes statements about *the next state*, not the long-term behavior.

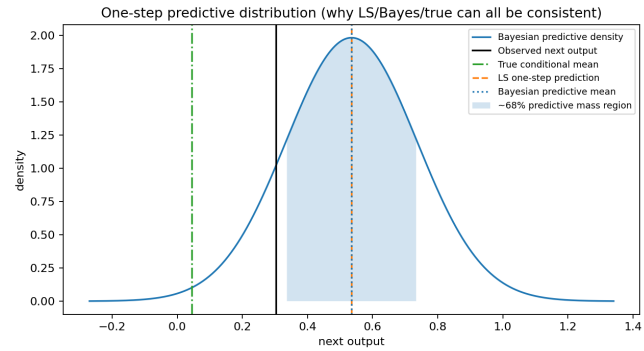

Fig. 1. Basic Bayesian predictive functionality. The predictive density places a distribution over the next output rather than only a point estimate. The observed next output, the least-squares one-step predictor, and the Bayesian predictive mean can differ while all remaining statistically consistent with the same underlying dynamics. The true conditional mean is the output of the true model used to generate the data.

III. Underconstrained and sub-Nyquist identification

Bayesian methods are most persuasive when the data are not informative enough to justify sharp nominal conclusions. Two such cases are especially important in controls.

Underconstrained identification. If $\Phi^\top\Phi$ is poorly conditioned, many parameter vectors yield similar one-step fit. Least squares selects one point on that ridge. The posterior instead represents the full family of plausible dynamics.

Sub-Nyquist or weakly sampled identification. If the data rate does not adequately resolve the underlying dynamics, several discrete-time models may agree on the sampled observations while disagreeing strongly on intersample or high-frequency behavior. In such cases, a nominal fit can look smooth precisely because aliasing and coarse sampling have hidden the ambiguity.

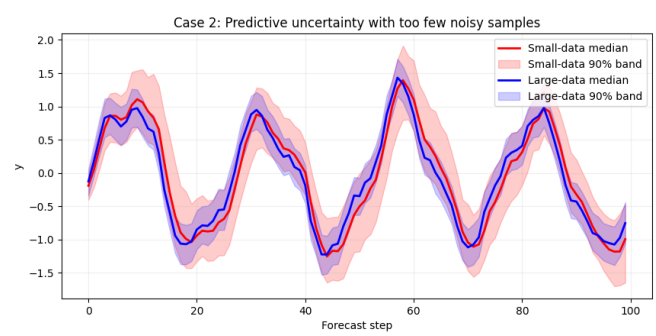

Fig. 2. Underconstrained identification in a small-data regime. The predictive medians from the small-data and large-data cases are similar, but the posterior band is much wider when only a few noisy samples are available. This is the operational advantage of Bayes: rather than returning one nominal fit with hidden fragility, it shows directly that the data support a broad family of trajectories.

Fig. 2 illustrates the underconstrained case. The posterior median may still follow the overall trend, but the predictive band reveals that many output sequences remain plausible. This is precisely the regime in which robust-control decisions

should not be based on a single estimated plant.

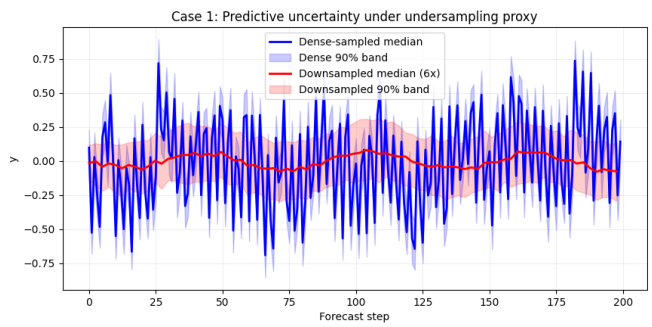

Fig. 3. Undersampling proxy for sub-Nyquist identification. Dense sampling yields a rapidly oscillatory posterior median, while downsampled data induce a smoother median that conceals unresolved fast dynamics. The posterior band widens under downsampling, correctly indicating that the apparent simplicity of the coarsely sampled trajectory is not evidence of stronger identification.

Fig. 3 shows the same point from a sampling perspective. Coarse or effectively sub-Nyquist-rate data can make the identified behavior look smoother while actually increasing uncertainty about the true fast dynamics. This is one of the clearest cases where Bayesian uncertainty quantification is not optional bookkeeping but a control-relevant safeguard.

IV. From posterior samples to robustness-relevant objects

Posterior samples can be pushed directly into the quantities that control engineers already inspect. Given draws $\theta^{(i)} \sim p(\theta \mid \mathcal{D})$, define the empirical stability probability

$$\hat{p}_{\mathrm{stab}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\Big\{\rho\big(A(\theta^{(i)})\big) < 1\Big\}, \tag{6}$$

where $A(\theta^{(i)})$ is the companion realization induced by the ARX denominator. This converts the binary statement "the estimated model is stable" into the more meaningful statement "most posterior-consistent models are stable" or "stability is uncertain."

Fig. 4 visualizes this interpretation directly. A pole cloud is a compact summary of how identification uncertainty maps into stability uncertainty.

Frequency-domain uncertainty is handled similarly. For each posterior sample one evaluates the transfer function $H(e^{j\omega};\theta^{(i)})$. A pointwise envelope then reveals where the data constrain the dynamics tightly and where they do not.

A particularly useful scalar summary is the posterior distribution of the Nyquist margin

$$d_{\mathrm{Nyq}}(\theta) := \min_{\omega\in[0,\pi]} \left|1 + L(e^{j\omega};\theta)\right|, \tag{7}$$

where L is the loop transfer function under a fixed controller. Small values of d_{Nyq} mean that the loop passes close to the critical point -1, so nominal gain- or phase-margin conclusions are fragile. The posterior of d_{Nyq} is therefore a data-driven proxy for how much robustness margin remains once identification uncertainty is included.

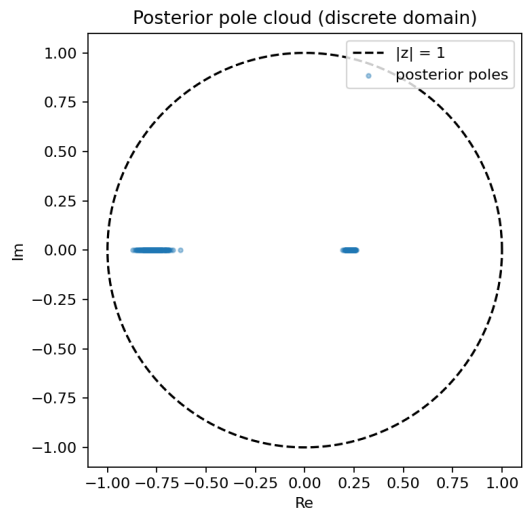

Fig. 4. Posterior pole cloud in the discrete-time z-plane. The unit circle is the stability boundary. A nominal stable estimate is far more convincing when posterior mass is concentrated well inside the unit circle than when it spreads toward the boundary.

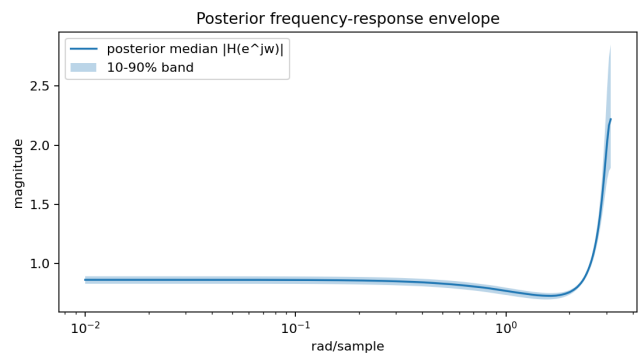

Fig. 5. Posterior frequency-response envelope. The low-frequency region is tight, whereas uncertainty grows near the higher-frequency end. For robust control, this indicates where nominal crossover or margin calculations are most susceptible to identification error.

Taken together, Figs. 4–6 suggest a practical viewpoint: the posterior is a data-driven ambiguity set, and the induced distributions of stability and Nyquist margin are interpretable surrogates for whether nominal robustness margins can be trusted.

V. Closed-loop propagation and posterior-risk control

The same sampling idea extends naturally to time-domain closed-loop performance. Fix a controller $C(z;\eta_c)$, a reference or disturbance signal, and an initial condition. For each posterior draw $\theta^{(i)}$, simulate the deterministic closed-loop trajectory and collect the resulting output $y_{\mathrm{cl}}^{(i)}(t)$. A *closed-loop Monte Carlo band* is simply the pointwise quantile envelope of the set $\{y_{\mathrm{cl}}^{(i)}(t)\}_{i=1}^{N}$. The spread of this band is therefore induced by model uncertainty, not by process-noise averaging.

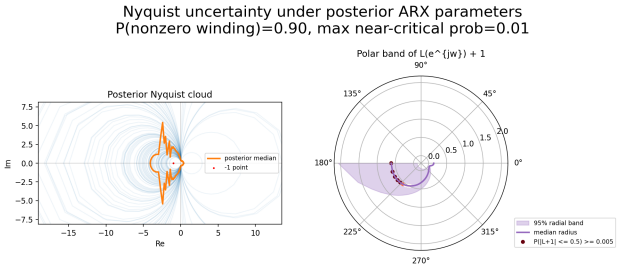

Fig. 6. Posterior Nyquist uncertainty plot. The left panel shows a cloud of posterior Nyquist curves; the right panel summarizes the radial distribution of $L(e^{j\omega})+1$. Robustness in SISO systems is shown as proximity of these loops to the critical point -1.

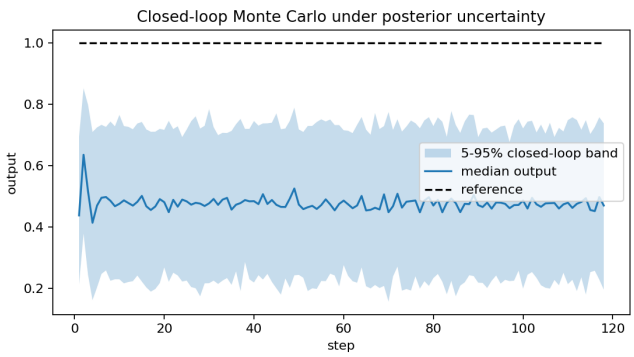

Fig. 7. Closed-loop Monte Carlo under posterior uncertainty. The median curve is the pointwise median across posterior-consistent closed-loop simulations, and the shaded band is a pointwise quantile band. The figure should be read as uncertainty in plant dynamics propagated through a fixed controller, not as repeated noisy time-series trials of one known plant.

Fig. 7 is important because robust-control claims are usually made in the closed loop, not in open-loop prediction space. A loop that looks acceptable at the posterior median may still have an uncomfortably wide family of transients once plant uncertainty is propagated.

This leads directly to posterior-risk design. Let $J_{\mathrm{cl}}(\eta_c, \theta)$ be a closed-loop performance cost, such as integrated tracking error plus control effort. A Bayesian robust-design template is

$$\begin{aligned} \eta_c^\star \in \arg\min_{\eta_c} \; & \mathrm{CVaR}_{1-\alpha}\big(J_{\mathrm{cl}}(\eta_c, \theta)\big) \\ \text{s.t. } & \mathbb{P}_{\theta\sim p(\theta|\mathcal{D})}\big(\rho\big(A_{\mathrm{cl}}(\eta_c, \theta)\big) < 1\big) \geq 1-\delta. \end{aligned} \tag{8}$$

Where $\mathrm{CVaR}_{1-\alpha}$ is the expected value of the α-proportion of the distribution in the upper tail, i.e. $\mathrm{CVaR}_{1-\alpha}(X) = \mathbb{E}[X|X \geq 1-\alpha \text{ quantile of } X]$. This is not yet full μ-synthesis, but it is already more principled than nominal tuning: the posterior acts as a probabilistically weighted uncertainty set learned from data.

VI. Posterior-wide destabilization via Δ-feedback

The attack-mitigation agenda enters naturally through a robust-control interconnection. Consider an augmented plant

$$\begin{bmatrix} z \\ y \\ r \end{bmatrix} = \begin{bmatrix} M & N \\ J & L \\ Q & S \end{bmatrix} \begin{bmatrix} w \\ d \end{bmatrix}, \qquad w = \Delta z, \tag{9}$$

where d is an exogenous disturbance or attack input, y is a damage channel, and r is a visibility channel such as an innovation residual. Eliminating w yields

$$T_{d\to y}(\Delta) = L + J(I - \Delta M)^{-1}\Delta N, \tag{10}$$

$$T_{d\to r}(\Delta) = S + Q(I - \Delta M)^{-1}\Delta N. \tag{11}$$

This is the Doyle–Glover–Khargonekar–Francis viewpoint on uncertainty and feedback interconnections [8], [9]. In the Bayesian setting, however, the plant blocks inherit posterior uncertainty from identification. One can therefore pose a posterior-wide destabilization problem

$$\Delta^\star \in \arg\max_{\Delta\in\mathcal{D}} \mathbb{P}_{\theta\sim p(\theta|\mathcal{D})}\big(\rho\big(A_{\mathrm{cl}}(\theta, \Delta)\big) \geq 1\big) - \lambda \, \|T_{d\to r}(\Delta)\|_\infty, \tag{12}$$

possibly with structural constraints on Δ. The first term seeks perturbations that destabilize a large fraction of posterior-consistent plants; the second penalizes visibility. This formulation is deliberately forward-looking, but it is technically natural: once one has a Bayesian plant ensemble, attack design and mitigation can be posed over that ensemble rather than over a single nominal model.

VII. Conclusion

Bayesian identification should be viewed as a robust-control tool, not merely a statistical embellishment. In underconstrained, weakly excited, or sub-Nyquist regimes, the central question is not which model fits best, but what family of plants remains compatible with the data and how that family affects robustness and security conclusions. Bayesian ARX offers a tractable answer: closed-form posteriors, interpretable priors, and uncertainty objects that map directly into poles, frequency response, Nyquist margin, and closed-loop transients.

The most immediate next steps are posterior-risk controller synthesis, observer design with posterior-aware residual shaping, and Bayesian mitigation of posterior-wide destabilizing perturbations. These are narrow enough to be technically credible and broad enough to matter in security-sensitive cyber-physical systems.

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